4.1 Experimental setup

We train a ResNet-18 on CIFAR-10 for 300 epochs with batch size 128. The learning rate is set to 3e-4 with a cosine decay schedule, and the weight decay is 0.05. Dropout is 0.1. All models are trained on 8 GPUs. The optimizer is AdamW. The random seed is not fixed in our released code.

4.2 Results

Figure 1 shows the training and validation curves over 300 epochs. Figure 2 reports the infrared spectrum of the sample, with the main absorption band at 1730 cm-1. Table 1 lists the hyperparameters used in all experiments.

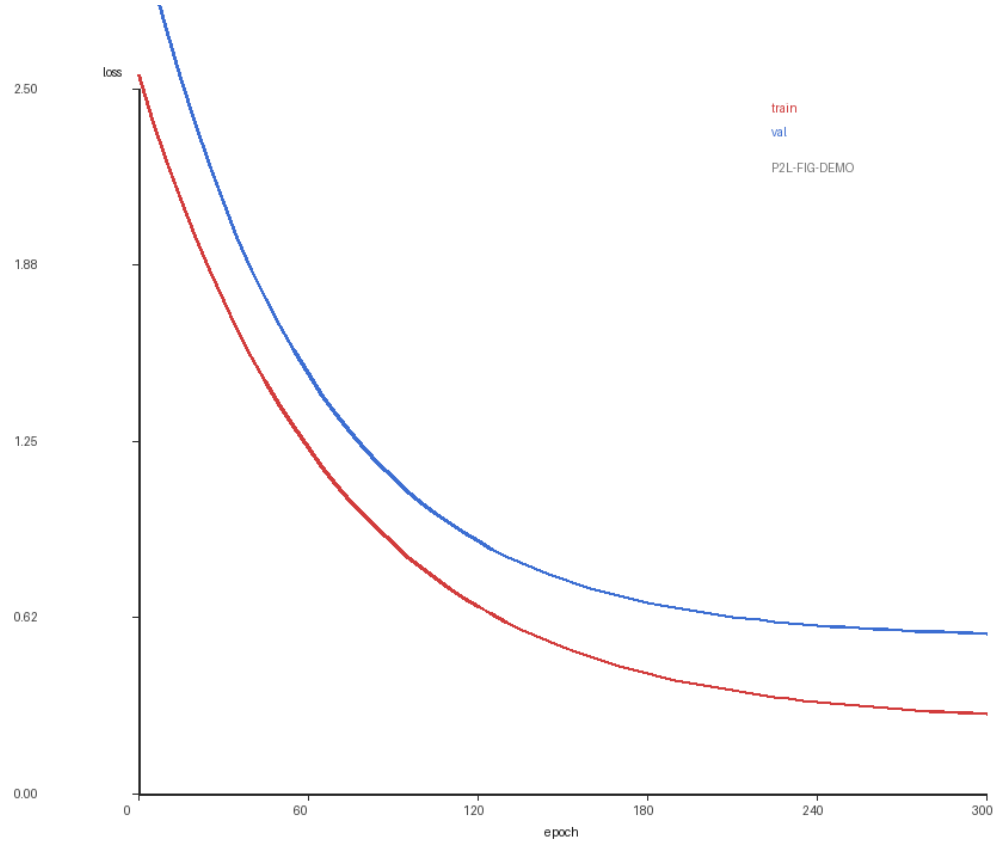


图 1 训练与验证损失曲线（300 epochs）

Table 1: Hyperparameters used in all experiments.

learning rate	3e-4
batch size	128
epochs	300
optimizer	AdamW
weight decay	0.05
dropout	0.1
lr scheduler	cosine

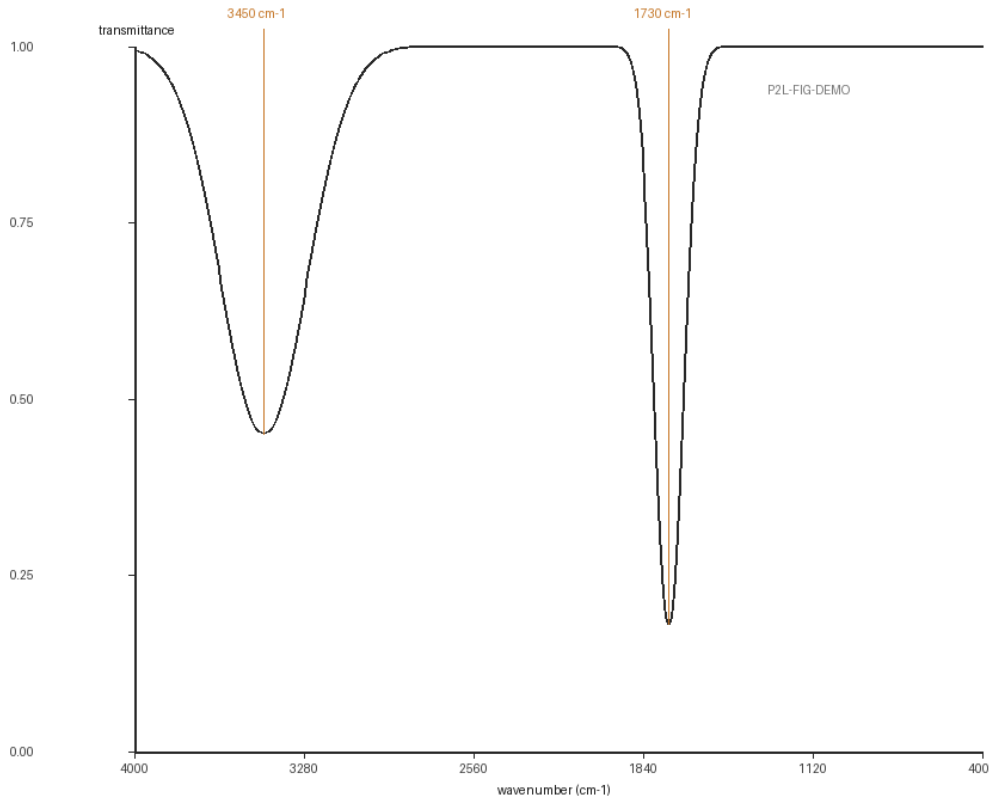


图 2 样品的红外光谱（KBr 压片），主吸收带位于 1730 cm-1